The Æther-flow Interpretation of Relativity

Primitive-Reservoir Observer-Reduction
Section-Centered Hidden-Response
Source-Balance Law
from a Centered Hidden-Response
Source-Penalty Kernel Law

Abstract

The previously live frontier theorem already proves that the current centered hidden-response residual-source law is the canonical response-orthogonal balanced-kernel law of a smaller centered hidden-response source-balance law on the fixed displacement fibers. The next honest benchmark-facing move is therefore no longer another replay of that source-balance law, of the residual-source law, of the centered residual law, of the hidden-response Hessian minimization law, of the square-root operator core, of the normalized-lift package, of the body-level forcing law, or of the visible-response structures that follow from them. It is to derive or justify the centered hidden-response source-balance law itself from still more primitive explicit substrate structure.

The present note takes that step in the smallest clean operator form now available below the centered-hidden-response-source-balance-law frontier. For each sector, it introduces a benchmark-faithful centered hidden-response source-penalty kernel law: the fixed proto-section benchmark base and fixed section-centered displacement body together with a finite-dimensional Euclidean balanced source carrier, a positive semidefinite self-adjoint source-penalty operator on that carrier, a smooth surjective source displacement readout to the actual displacement fibers, and a smooth positive self-adjoint source-response operator. The current centered hidden-response source-balance law is then no longer primitive. Its balance target is canonically the image of the positive square root of the source-penalty operator, and its balance operator is canonically that square root itself,

$$\mathcal{C}_\sigma^{\mathrm{disp,hid,res,src,bal}} = \operatorname{im} N_\sigma^{\mathrm{disp,hid,res,src,bal,pen}}, \qquad B_\sigma^{\mathrm{disp,hid,res,src,bal}} = N_\sigma^{\mathrm{disp,hid,res,src,bal,pen}}.$$

Equivalently, the current balanced kernel is exactly

$$\mathcal{Z}_\sigma^{\mathrm{disp,hid,res,src,bal}} = \ker B_\sigma^{\mathrm{disp,hid,res,src,bal}} = \ker C_\sigma^{\mathrm{disp,hid,res,src,bal,pen}},$$

so the benchmark-relevant content of the source-balance law is already forced by the smaller source-penalty kernel package.

The benchmark boundary remains unchanged throughout. The one operative metric is still exactly $g^\sharp$, the ordinary matter route is unchanged, there is no second operative metric, the low-energy matter law is not enlarged, and no stronger promotion from adoption to completed first-principles derivation is licensed. Once this smaller source-penalty kernel law is fixed, the same centered hidden-response source-balance law, the same centered hidden-response residual-source law, the same centered hidden-response residual law, the same hidden-response Hessian minimization law, the same square-root operator core, the same section-centered displacement-body quadratic forcing law, the same visible-response operator family, the same canonical minimal

visible-response representative, the same current body-response core, and every currently recorded downstream benchmark-facing consequence follow unchanged. The remaining honest burden therefore moves below the separate centered hidden-response source-balance law itself to a deeper derivation or justification of the centered hidden-response source-penalty kernel law that forces it.

1. Introduction

The active derivational program of Æther-flow Interpretation of Relativity has already moved one layer below the square-root-operator-core frontier, then below the hidden-response-Hessian frontier, then below the centered-residual frontier, and then below the centered-hidden-response-residual-source frontier itself. The previously live frontier theorem shows that the current centered hidden-response residual-source law is already the canonical response-orthogonal balanced-kernel law of a smaller centered hidden-response source-balance law on the fixed displacement fibers.

The next honest question is therefore narrower again. What still more primitive explicit substrate structure forces that centered hidden-response source-balance law itself? The present note answers that question in the smallest clean source-penalty form now available below the current frontier. It does not reinstall a separately primitive source-balance target or a separately primitive source-balance operator. Instead it keeps only a balanced source carrier, a positive semidefinite source-penalty operator whose kernel is the balanced state space, a source displacement readout, and a positive source-response operator. The current source-balance law then descends canonically because its balance target and balance operator are already forced by the positive square root of that source penalty.

Framework and claim boundary. Æther / Æther-flow framework denotes the broader Æther / Æther-flow framework built on the claims that the Æther is the underlying four-dimensional substrate of reality, the Æther-flow is its intrinsic ordered motion, observed three-dimensional space is the local experiential slice of that deeper substrate, S-time is the experienced order of change arising from matter, light, and the Æther-flow, and observed expansion is the three-dimensional appearance of deeper four-dimensional ordered motion. Within that framework, Æther-flow Interpretation of Relativity denotes the exact-closure theory in which the effective gravitational dynamics are adopted to be exactly Einsteinian with universal matter coupling. In the active sequence, *adoption* means use of that established relativistic dynamics without claiming substrate derivation, while *derivation* is reserved for a first-principles recovery from explicit substrate variables.

The present note stays strictly inside that boundary. It does not claim a completed first-principles derivation of GR from final substrate microphysics. Its narrower positive role is to show that the current centered hidden-response source-balance law already follows canonically from a smaller centered hidden-response source-penalty kernel law on the fixed displacement fibers.

2. Fixed Inputs from the Current Source-Balance Frontier

Definition 1 (Recorded primitive continuation data)**.** *Fix the following continuation data from the active primitive-reservoir observer-reduction line.*

1. *For each sector label* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$*, a primitive forcing shell*

$$\mathcal{I}_\sigma^{\mathrm{for}}.$$

2. *The product shell*

$$\mathcal{S}_{\mathrm{prim}} \equiv \mathcal{I}_{\mathrm{car}}^{\mathrm{for}} \times \mathcal{I}_{\mathrm{cur}}^{\mathrm{for}} \times \mathcal{I}_{\mathrm{hom}}^{\mathrm{for}}.$$

3. *The recorded metric and ordinary-matter outputs*

$$\mathscr{G}_{\mathrm{out}} : \mathcal{S}_{\mathrm{prim}} \to \{\textit{observer metrics}\}, \qquad \mathscr{T}_{\mathrm{out}} : \mathcal{S}_{\mathrm{prim}} \times \Psi \to \{\textit{observer stress tensors}\},$$

satisfying

$$\mathscr{G}_{\mathrm{out}}(\mathbf{w}) = g^\sharp, \qquad \mathscr{T}_{\mathrm{out}}(\mathbf{w}, \psi) = T_{\mu\nu}^{\mathrm{matter}}[g^\sharp, \psi] \tag{1}$$

for every $\mathbf{w} \in \mathcal{S}_{\mathrm{prim}}$ *and every matter configuration* $\psi \in \Psi$*.*

Definition 2 (Recorded benchmark base and downstream chain)**.** *Fix* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$*. The active route already fixes the following data and consequences.*

1. *the current proto-section benchmark base and in particular the recorded proto-section body* $\mathcal{B}_\sigma^{\mathrm{psec}}$ *and shell* $\mathcal{Z}_\sigma^{\mathrm{psec}}$*;*

2. *the fixed section-centered displacement body*

$$\mathcal{D}_\sigma^{\mathrm{pppresec}} \subset \mathcal{B}_\sigma^{\mathrm{psec}} \times V_\sigma^{\mathrm{pppresec}}$$

whose fiber

$$\mathcal{D}_\sigma^{\mathrm{pppresec}}(y) \equiv \{v \in V_\sigma^{\mathrm{pppresec}} : (y, v) \in \mathcal{D}_\sigma^{\mathrm{pppresec}}\}$$

is nonempty, compact, convex, contains 0*, and has nonempty interior for every* $y \in \mathcal{B}_\sigma^{\mathrm{psec}}$*;*

3. *the recorded downstream continuation*

$$\mathfrak{S}^{\mathrm{sub,proto,pppresec,pb,bodyresp}}, \quad \mathfrak{S}^{\mathrm{sub,proto,pppresec,pb,bodyop}}, \quad \mathfrak{R}^{\mathrm{sub,proto,presec,res}}, \quad \mathfrak{R}^{\mathrm{sub,proto,presec}},$$
$$\mathfrak{S}^{\mathrm{sub,proto,pppresec,pb,op}}, \quad \mathfrak{S}^{\mathrm{sub,proto,pppresec,pb}}, \quad \mathfrak{R}^{\mathrm{sub,proto,presec,sec}},$$

together with the current continuity-Hessian compressed core and benchmark one-metric observer package;

4. *benchmark faithfulness, meaning preservation of the benchmark outputs* (1)*, no second operative metric, no enlarged low-energy matter law, no change to the ordinary matter route, and no premature promotion from adoption to completed derivation.*

Definition 3 (Benchmark-faithful centered hidden-response source-balance law)**.** *Fix* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$*. A* benchmark-faithful centered hidden-response source-balance law *for sector* σ *consists of the following data.*

1. *The fixed proto-section benchmark base and fixed section-centered displacement body of Definition 2(1)–(2).*

2. *A finite-dimensional Euclidean balanced source carrier*

$$\mathcal{Y}_\sigma^{\mathrm{disp,hid,res,src,bal}}.$$

3. *A finite-dimensional Euclidean source-balance target*

$$\mathcal{C}_\sigma^{\mathrm{disp,hid,res,src,bal}}.$$

4. *A surjective linear source-balance operator*

$$B_\sigma^{\mathrm{disp,hid,res,src,bal}} : \mathcal{Y}_\sigma^{\mathrm{disp,hid,res,src,bal}} \to \mathcal{C}_\sigma^{\mathrm{disp,hid,res,src,bal}}.$$

5. *A smooth family of surjective linear source displacement readouts*

$$\Xi_\sigma^{\mathrm{disp,hid,res,src,bal}}(y) : \mathcal{Y}_\sigma^{\mathrm{disp,hid,res,src,bal}} \to V_\sigma^{\mathrm{pppresec}}, \qquad y \in \mathcal{B}_\sigma^{\mathrm{psec}}.$$

6. *A smooth family of positive self-adjoint source-response operators*

$$H_\sigma^{\mathrm{disp,hid,res,src,bal}}(y) : \mathcal{Y}_\sigma^{\mathrm{disp,hid,res,src,bal}} \to \mathcal{Y}_\sigma^{\mathrm{disp,hid,res,src,bal}}, \qquad y \in \mathcal{B}_\sigma^{\mathrm{psec}},$$

   each of which is positive definite on $\mathcal{Y}_\sigma^{\mathrm{disp,hid,res,src,bal}}$.

7. *Response-orthogonal balance corrections are readout-invisible:*

$$H_\sigma^{\mathrm{disp,hid,res,src,bal}}(y)^{-1} B_\sigma^{\mathrm{disp,hid,res,src,bal}*} \mathcal{C}_\sigma^{\mathrm{disp,hid,res,src,bal}} \subseteq \ker \Xi_\sigma^{\mathrm{disp,hid,res,src,bal}}(y) \qquad \text{for every } y \in \mathcal{B}_\sigma^{\mathrm{psec}}.$$

8. *Benchmark faithfulness in the sense of Definition 2(4).*

Definition 4 (Benchmark-faithful centered hidden-response source-penalty kernel law)**.** *Fix* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$. *A* benchmark-faithful centered hidden-response source-penalty kernel law *for sector* σ *consists of the following data.*

1. *The fixed proto-section benchmark base and fixed section-centered displacement body of Definition 2(1)–(2).*

2. *A finite-dimensional Euclidean balanced source carrier*

$$\mathcal{Y}_\sigma^{\mathrm{disp,hid,res,src,bal}}.$$

3. *A positive semidefinite self-adjoint source-penalty operator*

$$C_\sigma^{\mathrm{disp,hid,res,src,bal,pen}} : \mathcal{Y}_\sigma^{\mathrm{disp,hid,res,src,bal}} \to \mathcal{Y}_\sigma^{\mathrm{disp,hid,res,src,bal}}.$$

4. *A smooth family of surjective linear source displacement readouts*

$$\Xi_\sigma^{\mathrm{disp,hid,res,src,bal}}(y) : \mathcal{Y}_\sigma^{\mathrm{disp,hid,res,src,bal}} \to V_\sigma^{\mathrm{pppresec}}, \qquad y \in \mathcal{B}_\sigma^{\mathrm{psec}}.$$

5. *A smooth family of positive self-adjoint source-response operators*

$$H_\sigma^{\mathrm{disp,hid,res,src,bal}}(y) : \mathcal{Y}_\sigma^{\mathrm{disp,hid,res,src,bal}} \to \mathcal{Y}_\sigma^{\mathrm{disp,hid,res,src,bal}}, \qquad y \in \mathcal{B}_\sigma^{\mathrm{psec}},$$

   each of which is positive definite on $\mathcal{Y}_\sigma^{\mathrm{disp,hid,res,src,bal}}$.

6. *Let* $N_\sigma^{\mathrm{disp,hid,res,src,bal,pen}}$ *denote the unique positive square root of* $C_\sigma^{\mathrm{disp,hid,res,src,bal,pen}}$*. Its response-orthogonal penalty directions are readout-invisible:*

$$H_\sigma^{\mathrm{disp,hid,res,src,bal}}(y)^{-1}\,\mathrm{im}\,N_\sigma^{\mathrm{disp,hid,res,src,bal,pen}} \subseteq \ker \Xi_\sigma^{\mathrm{disp,hid,res,src,bal}}(y) \qquad \text{for every } y \in \mathcal{B}_\sigma^{\mathrm{psec}}.$$

7. *Benchmark faithfulness in the sense of Definition 2(4).*

Remark 1. *Definition 4 is strictly smaller at benchmark scope than Definition 3. It does not install a separately primitive balance target and does not install a separately primitive balance operator. Both will be forced canonically by the positive square root of the source-penalty operator. The benchmark-relevant balanced kernel is therefore already fixed by the smaller source-penalty datum itself.*

3. Canonical Source-Balance Law Forced by the Penalty Kernel

Definition 5 (Penalty kernel, induced target, and induced balance operator)**.** *Fix* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$ *and a benchmark-faithful centered hidden-response source-penalty kernel law. Define*

$$\mathcal{V}_\sigma^{\mathrm{disp,hid,res,src,bal,pen}} \equiv \ker C_\sigma^{\mathrm{disp,hid,res,src,bal,pen}}.$$

Let $N_\sigma^{\mathrm{disp,hid,res,src,bal,pen}}$ *be the unique positive square root of* $C_\sigma^{\mathrm{disp,hid,res,src,bal,pen}}$*. Define the induced source-balance target by*

$$\mathcal{C}_\sigma^{\mathrm{disp,hid,res,src,bal}} \equiv \mathrm{im}\,N_\sigma^{\mathrm{disp,hid,res,src,bal,pen}},$$

equipped with the Euclidean inner product inherited from $\mathcal{Y}_\sigma^{\mathrm{disp,hid,res,src,bal}}$*, and define the induced source-balance operator*

$$B_\sigma^{\mathrm{disp,hid,res,src,bal}} \equiv N_\sigma^{\mathrm{disp,hid,res,src,bal,pen}} : \mathcal{Y}_\sigma^{\mathrm{disp,hid,res,src,bal}} \to \mathcal{C}_\sigma^{\mathrm{disp,hid,res,src,bal}}.$$

Finally define

$$\mathcal{Z}_\sigma^{\mathrm{disp,hid,res,src,bal}} \equiv \ker B_\sigma^{\mathrm{disp,hid,res,src,bal}}.$$

Proposition 1 (The induced balance operator is surjective and has the penalty kernel as its kernel)**.** *For each sector* σ*, the induced source-balance operator of Definition 5 is surjective and satisfies*

$$\mathcal{Z}_\sigma^{\mathrm{disp,hid,res,src,bal}} = \ker B_\sigma^{\mathrm{disp,hid,res,src,bal}} = \ker C_\sigma^{\mathrm{disp,hid,res,src,bal,pen}} = \mathcal{V}_\sigma^{\mathrm{disp,hid,res,src,bal,pen}}.$$

Proof. Surjectivity is immediate from the definition of $\mathcal{C}_\sigma^{\mathrm{disp,hid,res,src,bal}}$ as the image of $N_\sigma^{\mathrm{disp,hid,res,src,bal,pen}}$.

Because

$$C_\sigma^{\mathrm{disp,hid,res,src,bal,pen}} = \left(N_\sigma^{\mathrm{disp,hid,res,src,bal,pen}}\right)^2$$

with $N_\sigma^{\mathrm{disp,hid,res,src,bal,pen}}$ self-adjoint, we have

$$\left\langle N_\sigma^{\mathrm{disp,hid,res,src,bal,pen}} w, N_\sigma^{\mathrm{disp,hid,res,src,bal,pen}} w \right\rangle = \left\langle w, C_\sigma^{\mathrm{disp,hid,res,src,bal,pen}} w \right\rangle$$

for every $w \in \mathcal{Y}_\sigma^{\mathrm{disp,hid,res,src,bal}}$. Hence

$$N_\sigma^{\mathrm{disp,hid,res,src,bal,pen}} w = 0 \iff \left\langle w, C_\sigma^{\mathrm{disp,hid,res,src,bal,pen}} w \right\rangle = 0 \iff C_\sigma^{\mathrm{disp,hid,res,src,bal,pen}} w = 0,$$

where the last equivalence uses the positive semidefiniteness and self-adjointness of $C_\sigma^{\mathrm{disp,hid,res,src,bal,pen}}$. Therefore the displayed kernels coincide, proving the claim. □

Theorem 1 (The source-penalty kernel law canonically forces the current source-balance law)**.** *For each sector $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$, any benchmark-faithful centered hidden-response source-penalty kernel law canonically determines a benchmark-faithful centered hidden-response source-balance law on the same fixed displacement fibers by taking the target and balance operator from Definition 5. In particular:*

1. *the balanced state space is exactly the penalty kernel,*

$$\mathcal{Z}_\sigma^{\mathrm{disp,hid,res,src,bal}} = \mathcal{V}_\sigma^{\mathrm{disp,hid,res,src,bal,pen}};$$

2. *the induced source-balance operator is canonical and surjective;*

3. *the readout invisibility condition required in Definition 3(7) is automatic from Definition 4(6).*

Proof. Items (1) and (2) follow from Proposition 1. It remains only to verify the source-balance-law axioms.

Items (1), (2), (5), (6), and (8) of Definition 3 are inherited directly from Definition 4. Item (3) holds because $\operatorname{im} N_\sigma^{\mathrm{disp,hid,res,src,bal,pen}}$ is a finite-dimensional Euclidean subspace of $\mathcal{Y}_\sigma^{\mathrm{disp,hid,res,src,bal}}$. Item (4) is exactly the induced operator of Definition 5, which is surjective by construction.

For item (7), because $N_\sigma^{\mathrm{disp,hid,res,src,bal,pen}}$ is self-adjoint and $\mathcal{C}_\sigma^{\mathrm{disp,hid,res,src,bal}} = \operatorname{im} N_\sigma^{\mathrm{disp,hid,res,src,bal,pen}}$ carries the inherited Euclidean inner product, we have

$$B_\sigma^{\mathrm{disp,hid,res,src,bal}*}\mathcal{C}_\sigma^{\mathrm{disp,hid,res,src,bal}} = \operatorname{im} N_\sigma^{\mathrm{disp,hid,res,src,bal,pen}}.$$

Therefore

$$H_\sigma^{\mathrm{disp,hid,res,src,bal}}(y)^{-1} B_\sigma^{\mathrm{disp,hid,res,src,bal}*}\mathcal{C}_\sigma^{\mathrm{disp,hid,res,src,bal}} = H_\sigma^{\mathrm{disp,hid,res,src,bal}}(y)^{-1} \operatorname{im} N_\sigma^{\mathrm{disp,hid,res,src,bal,pen}} \subseteq \ker \Xi_\sigma^{\mathrm{dis}}$$

for every $y \in \mathcal{B}_\sigma^{\mathrm{psec}}$ by Definition 4(6). Hence the induced package is a benchmark-faithful centered hidden-response source-balance law. □

Corollary 1 (The current centered hidden-response residual-source law is already forced one layer lower)**.** *Fix $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$. Any benchmark-faithful centered hidden-response source-penalty kernel law canonically determines the current centered hidden-response residual-source law on the fixed displacement fibers, and therefore also the current centered hidden-response residual law, the current hidden-response Hessian minimization law, the current square-root operator core, the current section-centered displacement-body quadratic forcing law, the visible-response operator family $S_\sigma^{\mathrm{disp,resp,vis}}$, and the canonical minimal visible-response representative $\mathfrak{S}^{\mathrm{disp,resp,red,min}}$.*

Proof. Theorem 1 provides the current centered hidden-response source-balance law canonically from the smaller source-penalty kernel package. The current source-balance-law frontier theorem then applies directly: once that source-balance law is fixed, the current centered hidden-response residual-source law is its canonical response-orthogonal balanced-kernel law. The already-recorded downstream continuation then yields the centered residual law, the hidden-response Hessian minimization law, the square-root operator core, the body-level forcing law, the visible-response operator

family, the canonical minimal visible-response representative, and the same current body-response core and downstream benchmark-facing consequences. □

Remark 2. *The present theorem counts as a genuine one-level gain below the current frontier because it removes two pieces of primitive source-balance data at once: the balance target and the balance operator. What remains primitive is only the carrier-level source penalty whose square root recreates them canonically.*

4. Benchmark Effect and Claim Boundary

Proposition 2 (Benchmark package unchanged)**.** *The source-penalty-kernel derivation leaves the benchmark package unchanged. In particular, the one operative metric remains exactly $g^\sharp$, the ordinary matter route remains the same, there is no second operative metric, the low-energy matter law is not enlarged, and no stronger promotion from adoption to completed first-principles derivation is licensed.*

Proof. Definitions 1, 2, and 4 already lock the benchmark outputs (1) and the claim boundary. Theorem 1 only repackages part of the primitive source data by inducing the balance target and balance operator from the positive source penalty. Corollary 1 then recovers the previously recorded downstream structures without changing observer-side output, matter coupling, or benchmark language. Hence the benchmark package is unchanged. □

5. Conclusion

The positive result of this note is that the current centered hidden-response source-balance law is no longer primitive at benchmark scope. It is already canonically forced by a smaller centered hidden-response source-penalty kernel law on the fixed displacement fibers: the balance target is the image of the positive square root of the source penalty, the balance operator is that square root, and the balanced kernel is exactly the penalty kernel.

The scope remains narrow and honest. The benchmark package is unchanged: the one operative metric is still exactly $g^\sharp$, the ordinary matter route is unchanged, there is no second operative metric, and the low-energy matter law is not enlarged. Nothing here upgrades adoption to derivation or licenses stronger promotion language.

The next open burden is therefore clear. The next benchmark-facing manuscript should derive or justify the current centered hidden-response source-penalty kernel law $\mathfrak{K}^{\mathrm{disp,hid,res,src,bal,pen}}$ itself from still more primitive explicit substrate structure, or derive an equivalent still smaller package that canonically yields it. Another same-centered-hidden-response-source-penalty-kernel-law restatement, another same-centered-hidden-response-source-balance-law restatement, another same-centered-hidden-response-residual-source-law restatement, another same-centered-hidden-response-residual-law restatement, another same hidden-response-minimization-law restatement, another same square-root-core restatement, another same body-law restatement, or another same visible-response restatement is not the clean next manuscript target at current scope.